Set
Wed 9 Sep 2026

Problem Set 3 — Functions

Session 3 · Domain, codomain, and undoing

BEFORE YOU START No question here is answered by a verdict alone. "Injective" is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since…" is. To *disprove* injectivity or surjectivity you need one concrete witness — to *prove* either, you need an argument covering every element.

Part A · Mechanics

1 IMAGES AND PREIMAGES

Let $f : \mathbb{Z} \to \mathbb{Z}$ be given by $f(n) = 3n - 2$. Compute $f(\{-1, 0, 2\})$, $f^{-1}(\{7\})$ and $f^{-1}(\{5\})$. Explain in one sentence why the third answer is what it is.

SOLUTION $f(\{-1, 0, 2\}) = \{-5, -2, 4\}$. For $f^{-1}(\{7\})$: $3n - 2 = 7$ gives $n = 3$, so the preimage is $\{3\}$. For $f^{-1}(\{5\})$: $3n - 2 = 5$ gives $n = \frac{7}{3}$, which is not an integer, so no element of the domain maps to 5 and $f^{-1}(\{5\}) = \varnothing$. A preimage is allowed to be empty — that is exactly what it means for 5 to lie in the codomain but outside the image.

2 IMAGE AGAINST CODOMAIN

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 - 4x + 7$. State the image of f and prove your answer. Is f surjective? Justify.

SOLUTION Complete the square: $f(x) = (x - 2)^2 + 3$. Since $(x - 2)^2 \geq 0$ for every real x, we get $f(x) \geq 3$, so the image is contained in $[3, \infty)$. Conversely every $y \geq 3$ is attained, at $x = 2 + \sqrt{y - 3}$. Hence the image is exactly $[3, \infty)$. f is **not** surjective: the codomain is $\mathbb{R}$, and 0 (for instance) is never attained because $f(x) \geq 3 > 0$ for all x.

3 UNIT CIRCLE

Derive each value from the unit circle. For each, state the quadrant and the reference angle before you give the number. A recalled value without that working is not a derivation.

a. $\cos \dfrac{2\pi}{3}$

b. $\sin \dfrac{3\pi}{4}$

c. $\cos \dfrac{7\pi}{6}$

d. $\sin \dfrac{5\pi}{3}$

e. $\tan \dfrac{3\pi}{4}$

SOLUTION (a) $\frac{2\pi}{3}$ is in quadrant II, reference angle $\frac{\pi}{3}$; cosine is the x-coordinate, negative there, so $-\frac{1}{2}$. (b) $\frac{3\pi}{4}$, quadrant II, reference $\frac{\pi}{4}$; sine is the y-coordinate, positive there, so $\frac{\sqrt{2}}{2}$. (c) $\frac{7\pi}{6}$, quadrant III, reference $\frac{\pi}{6}$; cosine negative, so $-\frac{\sqrt{3}}{2}$. (d) $\frac{5\pi}{3}$, quadrant IV, reference $\frac{\pi}{3}$; sine negative, so $-\frac{\sqrt{3}}{2}$. (e) $\tan \frac{3\pi}{4} = \dfrac{\sin(3\pi/4)}{\cos(3\pi/4)} = \dfrac{\sqrt{2}/2}{-\sqrt{2}/2} = -1$.

4 CLASSIFICATION

For each function decide whether it is injective, surjective, both or neither. Prove each positive claim; disprove each negative one with an explicit witness.

a. $f : \mathbb{R} \to \mathbb{R},\ f(x) = 5 - 3x$
b. $f : \mathbb{Z} \to \mathbb{Z},\ f(n) = n^2 - n$
c. $f : \mathbb{R} \to \mathbb{R},\ f(x) = x^3 - 3x$
d. $f : \mathbb{N} \to \mathbb{N},\ f(n) = n^2$
e. $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\},\ f(x) = \dfrac{1}{x}$
f. $f : \mathbb{R} \to [-1, 1],\ f(x) = \sin x$
g. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},\ f(a, b) = 2a + 4b$
h. $S = \{1, 2, 3\},\ f : \mathcal{P}(S) \to \mathcal{P}(S),\ f(X) = S \setminus X$

SOLUTION (a) **Bijective.** Injective: $5 - 3a = 5 - 3b \Rightarrow a = b$. Surjective: given y, take $x = \frac{5-y}{3}$.
(b) **Neither.** Not injective: $f(0) = f(1) = 0$. Not surjective: $n^2 - n = n(n-1)$ is a product of consecutive integers, hence even, so the odd value 1 is never attained.
(c) **Surjective, not injective.** Not injective: $f(0) = 0$ and $f(\sqrt{3}) = 3\sqrt{3} - 3\sqrt{3} = 0$. Surjective: f is continuous with $f(x) \to \pm\infty$ as $x \to \pm\infty$, so by the intermediate value theorem every real is attained.
(d) **Injective, not surjective.** Injective: for $a, b \geq 1$, $a^2 = b^2 \Rightarrow a = b$. Not surjective: 2 is not a perfect square.
(e) **Bijective.** $f(f(x)) = x$ on the whole domain, so f is its own two-sided inverse, and by the theorem a map with a two-sided inverse is a bijection.
(f) **Surjective, not injective.** Not injective: $\sin 0 = \sin \pi = 0$. Surjective: the image of $\sin$ is exactly $[-1, 1]$, which here *is* the codomain — note how shrinking the codomain created surjectivity.
(g) **Neither.** Not injective: $f(2, 0) = 4 = f(0, 1)$. Not surjective: $2a + 4b$ is always even, so 1 is never attained.
(h) **Bijective.** $f(f(X)) = S \setminus (S \setminus X) = X$, so f is its own inverse, hence a bijection.

5 COMPOSITION

Let $f : A \to B$ and $g : B \to C$.

a. Prove that if $g \circ f$ is surjective, then g is surjective.

b. Show by explicit example that f need *not* be surjective under the same hypothesis.

c. In one sentence, say why (a) is about g and not f, given that in class we proved $g \circ f$ injective forces f injective.

> **SOLUTION** (a) Let $c \in C$. Since $g \circ f$ is surjective there is $a \in A$ with $g(f(a)) = c$. Put $b = f(a) \in B$; then $g(b) = c$. As c was arbitrary, g is surjective.
> (b) $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, with $f(1) = 1$ and $g(1) = g(2) = 1$. Then $g \circ f$ is surjective onto C, but f misses 2.
> (c) Surjectivity is a statement about what comes *out*, so it is inherited by the last map in the chain; injectivity is a statement about what goes *in*, so it is inherited by the first.

6 RESTRICTION

Take the rule $x \mapsto x^2 - 4x + 7$ from question 2. Choose a domain $A \subseteq \mathbb{R}$ and a codomain B making $f : A \to B$ a **bijection**, and prove both halves.

> **SOLUTION** Take $A = [2, \infty)$ and $B = [3, \infty)$; write $f(x) = (x-2)^2 + 3$. Injective: for $x_1, x_2 \geq 2$ the quantities $x_1 - 2, x_2 - 2$ are ≥ 0, and $(x_1 - 2)^2 = (x_2 - 2)^2$ with both bases non-negative gives $x_1 - 2 = x_2 - 2$. Surjective: given $y \geq 3$, put $x = 2 + \sqrt{y - 3}$, which lies in A and satisfies $f(x) = y$. ($A = (-\infty, 2]$ with the same B works equally well.)

Part C · Communication

7 FIND THE ERROR

The following was produced by a chatbot. It contains **three** distinct errors. Find all three; for each, say what is wrong and give a witness where one is needed.

> **Claim.** $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x}{x^2+1}$, is a bijection, and its inverse is $f^{-1}(x) = \dfrac{x^2+1}{x}$.
> **"Proof."** f is injective because it is a ratio of polynomials with no repeated factors. It is surjective because its domain and codomain are both $\mathbb{R}$. Inverting a fraction gives the inverse function.

SOLUTION **Error 1 — not injective.** "Ratio of polynomials with no repeated factors" is not a criterion for anything. Witness: $f(2) = \frac{2}{5}$ and $f\left(\frac{1}{2}\right) = \frac{1/2}{5/4} = \frac{2}{5}$, so $f(2) = f\left(\frac{1}{2}\right)$ with $2 \neq \frac{1}{2}$.

Error 2 — not surjective. Domain and codomain being equal says nothing about surjectivity. In fact $(x \mp 1)^2 \geq 0$ gives $x^2 + 1 \geq \pm 2x$, so $|f(x)| \leq \frac{1}{2}$ for every x; the image is $\left[-\frac{1}{2}, \frac{1}{2}\right]$ and 1 is never attained.

Error 3 — inverse against reciprocal. $\frac{x^2+1}{x}$ is $\frac{1}{f(x)}$, not f^{-1}. Undoing a function is unrelated to flipping a fraction: f^{-1} satisfies $f^{-1}(f(x)) = x$, and here $\frac{1}{f(f(x))} \neq x$ in general. Since f is not a bijection, no inverse function exists at all.

8 REWRITE PRECISELY

Each sentence is sloppy or false as written. Rewrite each so that it is unambiguous and true.

a. "x^2 is not injective."

b. "Every function has an inverse — you just swap x and y."

c. "$\sin^{-1}(x)$ means $\dfrac{1}{\sin x}$."

WHAT A GOOD ANSWER DOES (a) A rule is not a function: e.g. "the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, is not injective, since $f(-1) = f(1)$ but $-1 \neq 1$." Any answer that supplies a domain and codomain and a witness is acceptable.
(b) "A function has an inverse if and only if it is a bijection. Swapping x and y in the equation of the graph yields a function precisely in that case."
(c) "$\sin^{-1}$ denotes the inverse of $\sin$ restricted to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, where it is a bijection onto $[-1, 1]$. The reciprocal $\frac{1}{\sin x}$ is a different function, written $\csc x$."